Ryne Shelton

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Professional Summary

Electrical & Computer Engineering professional (4.0 GPA) specializing in power distribution systems, interconnection compliance, transformer loading analysis, and grid event automation. Conducted multi-year PGE-funded research under Professor Robert Bass and USGS-funded machine learning research under Professor John Lipor, developing automated grid event response models and high-dimensional geospatial/electrical data pipelines. Skilled in validating single-line diagrams, assessing voltage impacts, performing initial technical screens, and producing clear technical documentation. Strong electrical CAD proficiency (Revit, Fusion 360, LTspice, BIM workflows) combined with advanced DSP, ML, and automated testing expertise. Familiar with NEC guidelines, IEEE distribution standards, and collaborative utility engineering workflows.

Core Electrical Engineering Competencies

Distribution System Design | Transformer Loading Analysis | Voltage Drop & Impact Studies

Interconnection Requirements | Single-Line Diagram Validation | Grid Event Automation

Electrical CAD Drafting (Revit, Fusion 360, LTspice, BIM Workflows)

Power System Data Modeling | ML for Grid Resilience | Technical Documentation

Technical Skills

Programming & Automation: Python, C++, MATLAB, NumPy, Pandas, Linux, Git, Shell Scripting.
Signal & ML Systems: CNNs, LSTM Filtering, TensorFlow, Predictive Modeling, Anomaly Detection.
Electrical CAD Tools: Revit, Fusion 360, LTspice, BIM workflows.
Instrumentation: Oscilloscopes, Digital Logic Analyzers, Sensor Data Analytics.
Systems Testing: Root-Cause Failure Analysis, Automated Calibration Scripts, Cleanroom Operations.

Selected Projects

Automated Grid Event Response System — ML-driven secondary system deployment reducing cascading failure risk for PGE distribution feeders.

Distribution Feeder Analysis Tool — Python-based transformer loading and voltage impact calculator supporting interconnection review.

USGS Geospatial Anomaly Detection Pipeline — High-dimensional ML system processing millions of sensor datapoints for reliability analysis.

Professional Experience

Post-silicon Validation Engineer — Jan 2025 – Aug 2025
Intel Corporation — Hillsboro, OR

- Executed root-cause failure analysis using SEM and FIB instrumentation.
- Isolated structural, electrical, and physical fault mechanisms on advanced node technologies.

Field Service Engineer — Feb 2024 – Jan 2025
Hitachi High-Tech America — Hillsboro, OR

- Calibrated and diagnosed CD-SEM metrology systems inside HVM cleanrooms.
- Used oscilloscopes and logic analyzers to isolate electromechanical signal discrepancies.
- Developed script-based validation protocols ensuring equipment matching and yield compliance.

Power Systems & Machine Learning Research Engineer — Sept 2021 – Aug 2024
Portland State University — Power Engineering Lab & USGS Research Group — Portland, OR

- Conducted joint PGE-funded and USGS-funded research under Professors Robert Bass and John Lipor.
- Analyzed transformer loading, service panel capacity, and voltage impacts across 15+ distribution feeders.
- Performed initial technical screens, validated 120+ single-line diagrams, and ensured interconnection compliance.
- Developed ML-driven automated grid event response models reducing event classification latency by 35%.
- Built high-dimensional geospatial/electrical data pipelines supporting USGS anomaly detection.
- Collaborated with multidisciplinary engineering teams and produced documentation for PGE and USGS stakeholders.

Intern Engineer — Jan 2022 – Sept 2022
Datalogic — Eugene, OR

- Designed CNN models for high-speed Data Matrix and QR code localization.
- Implemented LSTM-based noise reduction algorithms for robust signal validation.

Education & Leadership

Portland State University — Portland, OR
B.S. Electrical & Computer Engineering — MCECS Learning Center Manager — 2021–2024

Portland Community College — Portland, OR
Associate of Science | Lead Tutor & STEM Center Liaison — 2019–2021

Professional Interests

Grid Modernization & Resilience Engineering

Distribution Automation & Interconnection Engineering

Power System Modeling & Electrical CAD Design

Machine Learning for Energy Systems

Affiliations & Honors

Affiliations: IEEE | Computer Action Team (MCECS) | PSU Student Government Senator

Honors: NASA Aerospace Scholar | Louis Stokes Alliance Scholar | Renaissance Scholar